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PRODUCT REQUIREMENTS DOCUMENT

PROJECT:

Metering Management Tool

[ID]		C1

DOCUMENT DESCRIPTION:

DIGITAL R&M HUB

METERING MANAGEMENT TOOL

PRODUCT REQUIREMENTS DOCUMENT

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REVISION	REVISED CHAPTERS	REVISION DESCRIPTION	REASON FOR REVISION
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1. PURPOSE OF DOCUMENT

This section establishes the foundational context for the Metering Management Tool (MMT), defining the core problem being solved, the scope of the intended solution, and the primary audience for this Product Requirements Document.

1.1 PROBLEM CONTEXT

Flow measurement on offshore production facilities plays a critical role in operational control, production performance tracking, contractual compliance, and environmental emission monitoring. For units operating under Brazilian jurisdiction, the regulatory framework imposed by the ANP (Agência Nacional do Petróleo) — particularly through the Metrological Technical Regulation RTM 01/2013 (Resolução Conjunta ANP/INMETRO nº 1), Resolução ANP nº 18/2014 (Failure Notification), and Resolução ANP nº 52/2013 (Gas and Oil Sampling) — demands rigorous, auditable management of the entire metering system.

The absence of a centralized tool capable of capturing the nuances of local regulation and storing all relevant data in a single, auditable repository exposes the organization to significant risks: non-compliance fines, contractual disputes with clients, and loss of operational visibility across a fleet of 10 FPSOs.

1.2 PRODUCT SCOPE

The Metering Management Tool (MMT) is a web-based application designed to manage the complete lifecycle of fiscal and operational metering systems aboard SBM Offshore's FPSO fleet in Brazil. The solution is structured around 12 functional modules, as defined in the Functional Specification (IS93502 — OOSDOMPF550001, Rev. C1):

#	Module	Scope
M1	Equipment Registration & Installation	Serial number tracking, tag assignments, installation/removal history
M2	Metrological Confirmation	Calibration lifecycle, certificates, uncertainty calculations, FC updates
M3	Chemical Analysis	Periodic sampling workflows (CRO, PVT, Enxofre, Viscosity), lab report validation, SLA tracking
M4	Onshore Maintenance	Equipment maintenance tracking at onshore facilities
M5	Synchronize Data	Data exchange with external systems (ANP, IFS)
M6	Monitoring & Alerts	SLA-based alerting, overdue detection, compliance dashboards
M7	Export Data	Structured data export for ANP audits (XML, ZIP archives)
M8	Planning	Unified view of all upcoming calibrations, inspections, and samplings

M9	Failure Notification (NFSM)	Mismeasurement notification management per Resolução ANP nº 18/2014
M10	Historical Reports	Audit-ready report generation with full traceability
M11	Asset Hierarchy Browser (Configuration)	FPSO hierarchy modeling, metering system configuration, properties management
M12	User & Access Management	Role-based access control, authentication, audit logging

1.3 DOCUMENT PURPOSE

This Product Requirements Document (PRD) serves as the **single source of truth** for what the MMT must deliver and why. It translates the business needs captured in the Functional Specification into structured, traceable product requirements that guide all downstream activities — design, development, testing, and acceptance.

Specifically, this document:

1. **Defines business context and value**: Establishing the regulatory, operational, and contractual drivers that justify the initiative and the measurable outcomes expected.
2. **Specifies functional and non-functional requirements**: Detailing what the system must do (capabilities per module) and how it must perform (availability, security, scalability, compliance).
3. **Establishes traceability**: Linking each requirement to its originating business driver, functional specification reference, and acceptance criteria, enabling clear validation during User Acceptance Testing (UAT).
4. **Captures assumptions, constraints, and dependencies**: Documenting the boundaries within which the solution must operate, including regulatory constraints, data dependencies, and integration requirements with external systems.
5. **Serves as the contractual baseline**: Providing the agreed-upon requirements against which the contractor's deliverables will be evaluated, accepted, or rejected.

1.4 INTENDED AUDIENCE

Role	Usage
Metering Lead (ML) / Metering Engineer (ME)	Validate that requirements reflect operational reality
Product Owner / Business Analyst	Prioritize backlogs and trace features to business value
Development Team (Contractor)	Design and implement the solution per stated requirements
QA / UAT Team	Develop test cases and acceptance criteria
SBM Management / Stakeholders	Assess alignment with PACE program strategy and investment rationale
ANP Auditors (indirect)	Verify that the tool supports regulatory compliance obligations

1.5 DOCUMENT LIFECYCLE

This PRD is a **living document** maintained throughout the project lifecycle. It will be updated to reflect requirement changes approved through the project's change control process. All revisions will be tracked in the Revision Status table at the front of the document, with clear indication of what changed, who approved it, and the effective date.

The PRD complements, but does not replace, the detailed Software Requirements Specification (SRS) to be delivered by the contractor, which will provide implementation-level detail derived from the requirements stated herein.

2 REFERENCE DOCUMENTS

This section enumerates the external regulatory frameworks, international standards, and internal SBM specifications that govern the requirements and technical limits of the MMT project.

[1] Resolução Conjunta ANP/INMETRO nº 1 — RTM 01/2013
[2] Resolução ANP nº 18/2014 — Failure Notification
[3] Resolução ANP nº 52/2013 — Gas and Oil Sampling
[4] Resolução ANP nº 65/2014 — Data Exchange
[5] Ofício Circular ANP/NFP nº 01/2020 — RTM Clarifications
[6] ANP/NFP nº 02/2020, 835/2020, 841/2021 — COVID-19 Waivers
[7] Ofício Circular ANP/NFP nº 02/2021 — Resolução 18 Clarifications
[8] Ofício ANP/NFP nº 906/2021 — Offloading XML
[9] ISO 10012:2003 — Measurement management systems — Requirements for measurement processes and measuring equipment
[10] ISO 5167:2003 — Measurement of fluid flow by means of pressure differential devices
[11] Metering Management Tool — Functional Specification (IS93502, OOSDOMPF550001, Rev. C1)

3 ABBREVIATIONS AND DEFINITIONS

This section provides a standardized glossary of domain-specific terms and acronyms used throughout this document, ensuring unambiguous communication between business and technical teams.

3.1 ABBREVIATIONS

Term/Acronym	Definition	Context/Notes	Owner
ANP	National Petroleum Agency (Agência Nacional do Petróleo)	Brazilian regulatory body overseeing oil & gas measurement. Defines calibration frequencies, sampling requirements, and failure notification obligations. Non-compliance results in significant fines.	Regulatory / Metering Lead
API	Application Programming Interface	All MMT functionalities are exposed via REST APIs, enabling data access by SBM users and integration with external systems (Power BI, IFS).	Development Team
BSW	Basic Sediment and Water	Percentage of water and sediment in crude oil samples. Tracked as an operational analysis type in M3.	Metering Engineer
CA	Critical Analysis	Automated evaluation of a calibration certificate or lab report against predefined rules and historical data to detect anomalies (2σ check). Applies to M2 (certificates) and M3 (lab reports).	Metering Engineer
CRO	Chromatography (Cromatografia)	Gas composition analysis determining mole fractions, relative density, and heating value. Most frequent periodic analysis type (30/90/180-day intervals).	Laboratory / ME
CT	Custody Transfer	Metering classification for fiscal measurement points where oil/gas ownership changes. Highest regulatory scrutiny and shortest calibration intervals.	Metering Lead
FC	Flow Computer	Device that calculates flow rates from raw sensor inputs. FC updates are triggered after approved lab reports (CRO) with SLA tracking in business days.	Metering Technician
FPSO	Floating Production, Storage and Offloading	Offshore production vessel. The MMT manages metering systems across a fleet of 10 FPSOs operating in Brazil. Each FPSO is the top level of the asset hierarchy.	Operations
IFS	CMMS — Maintenance Management Software	SBM's enterprise maintenance system. MMT integrates with IFS for spare parts tracking and work order synchronization (M5).	IT / Maintenance
MA	Metering Analyst	Onshore role responsible for data analysis, compliance reporting, and ANP submission coordination.	Metering Team
ME	Metering Engineer	Onshore engineering role responsible for calibration planning, certificate validation, uncertainty calculations, and regulatory compliance. Primary MMT user for M2, M3, M8.	Metering Team
ML	Metering Lead	Senior role overseeing all metering activities across the FPSO fleet. Responsible for SLA	Metering Team

		compliance, audit readiness, and resource allocation.	
MMT	Metering Management Tool	The web-based application specified in this PRD. Central system for managing metering equipment, calibrations, chemical analyses, and regulatory compliance.	Product Owner
MT	Metering Technician	Offshore role executing calibrations, equipment installations/removals, seal management, and sample collection aboard FPSOs.	Metering Team
NFSM	Mismeasurement Notification of the Metering System (Notificação de Falha do Sistema de Medição)	Mandatory ANP notification issued when a measurement failure is detected. Tracked in M9 with strict regulatory timelines for submission.	Metering Engineer
P&ID	Piping and Instrumentation Diagram	Engineering drawing showing the process flow and instrument locations. Tags referenced in MMT correspond to instrument positions on P&IDs.	Engineering
PVT	Pressure-Volume-Temperature	Laboratory analysis determining fluid properties (density, RS, FE) under reservoir conditions. 90-day interval, requires validation, no FC update.	Laboratory / ME
RTM	Metrological Technical Regulation (Regulamento Técnico de Medição)	ANP/INMETRO regulation (RTM 01/2013) defining the metrological confirmation framework. Core regulatory reference for M2 and M4.	Regulatory
SLA	Service Level Agreement	Time-based targets for each phase of the chemical analysis workflow (disembark, laboratory, report, FC update). Configured in the SLA Matrix per classification, type, and location.	Metering Lead
SRS	Software Requirements Specification	Detailed technical specification derived from this PRD, to be produced by the development contractor.	Development Team
UAT	User Acceptance Testing	Formal testing phase where SBM end users validate that the delivered software meets the requirements stated in this PRD.	QA / Metering Team
UC	Uncertainty Calculation	Mathematical determination of measurement uncertainty per ISO standards. Generated within M2 after certificate issuance and subject to Critical Analysis.	Metering Engineer

3.2 DOMAIN DEFINITIONS

Term	Definition	Context/Notes	Owner
Metrological Confirmation	Set of operations required to ensure that measuring equipment conforms to the requirements for its intended use — including calibration, verification, adjustment, repair, recalibration, and sealing (per ISO 10012:2003).	Full lifecycle managed in M2, from planning through FC update and closure.	Metering Engineer
Critical Analysis (CA)	Automated evaluation of a report (certificate or lab result) against predefined acceptance criteria and historical data from the same equipment/sample point. Uses statistical rules (e.g., 2σ deviation check) to flag anomalies.	Applied in M2 (calibration certificates) and M3 (CRO/PVT lab reports). Flags reports as Approved or Reproved.	Metering Engineer
Tag (Instrument Tag)	Unique identifier of a physical location on the process plant where an instrument is installed. Tags are fixed addresses tied to P&ID positions; they do not change when equipment is replaced.	Tags define where an instrument sits. Certificates and processes are linked to the serial number (the what), with the tag providing location context.	Engineering
Serial Number	Unique identifier of a physical device (transmitter, orifice plate, flowmeter). Certificates are issued to serial numbers. A serial number may move between tags over its lifecycle.	Equipment module (M1) tracks all serial numbers, their specifications, current/historical tag assignments, and associated certificates.	Metering Technician
Emergency Sample	An unscheduled sample triggered when a periodic analysis is reproved during validation. Planned for the earlier of: 3 business days after report emission, or the next periodic collection date.	Ensures rapid re-analysis when lab results fail Critical Analysis. Only applies to reproved Fiscal/Apropriation CRO and PVT analyses.	Metering Engineer
SLA Matrix	Configuration table mapping each combination of Classification × Analysis Type × Location (Onshore/Offshore) to specific time-based targets: sampling interval, disembark, laboratory, report, and FC update deadlines.	Source of truth for the chemical analysis module (M3) dashboard signalling, due dates, and urgency indicators. Defined in the SLA Matrix.	Metering Lead
Override	Temporary replacement of a flow computer reading with a calculated reference value during calibration, as required by Resolução ANP nº 18/2014.	MMT must calculate and provide the correct override value to the MT. Duration and values must be logged for NFSM purposes.	Metering Technician

Offloading	Transfer of produced oil from the FPSO to a shuttle tanker. Requires specific XML data submission to ANP including metering readings and associated documentation.	Data structure defined by Ofício ANP/NFP nº 906/2021. Export handled by M7.	Operations / MA

4 CURRENT OVERVIEW & CONTEXT

This section maps the current "As-Is" operational landscape of SBM's metering processes, quantifying the fleet scale, existing reporting workflows, and historical regulatory exposures that necessitate a digital transformation.

4.1 BUSINESS ENVIRONMENT

SBM Offshore operates a fleet of FPSOs in Brazil under contracts with major oil companies (Petrobras, Shell). Flow measurement on these production facilities is critical across four fronts: **operational control**, **production performance**, **contractual targets**, and **environmental emission monitoring**.

For units operating under Brazilian jurisdiction, the **regulatory front** becomes the most critical. The ANP (National Petroleum Agency) and INMETRO enforce tight metrological regulations through RTM 01/2013, with non-compliance resulting in significant financial exposure.

4.2 OPERATIONAL SCALE

The metering management scope handled daily across Region 1 (Brazil) is substantial:

Metric	Value
FPSOs in operation	9 (+ 3 new units: Almirante Tamandaré, Alexandre de Gusmão, Sepetiba)
Regulated instruments	2,900
Flow computers	212
Critical parameters evaluated daily	> 1,000,000
Daily reports and parameter files	> 400
Calibration data points per month	360+
ANP XML file submissions per month	1,000+
Gas/oil analysis reports per month	80+
Calibration certificates per month	360+
Uncertainty calculations per month	120+
Audit trail evidences per month	500+

4.3 TEAM STRUCTURE

Role	Headcount	Location	Responsibilities
Metering Lead (ML)	1	Onshore	Fleet-wide compliance oversight, resource allocation, client interface
Metering Engineers (ME)	7	Onshore	Calibration planning, certificate validation, UC calculations, ANP reporting
Metering Analyst (MA)	1	Onshore	Data analysis, report consolidation, compliance monitoring
Metering Technicians (MT)	22	Offshore	Calibration execution, equipment installations, seal management, sample collection

The team reports to the Metering Lead, Maintenance Manager, and Asset Integrity Manager.

4.4 CURRENT INFORMATION FLOW (AS-IS)

Today, metering data flows between offshore and onshore through a predominantly manual, file-based process:

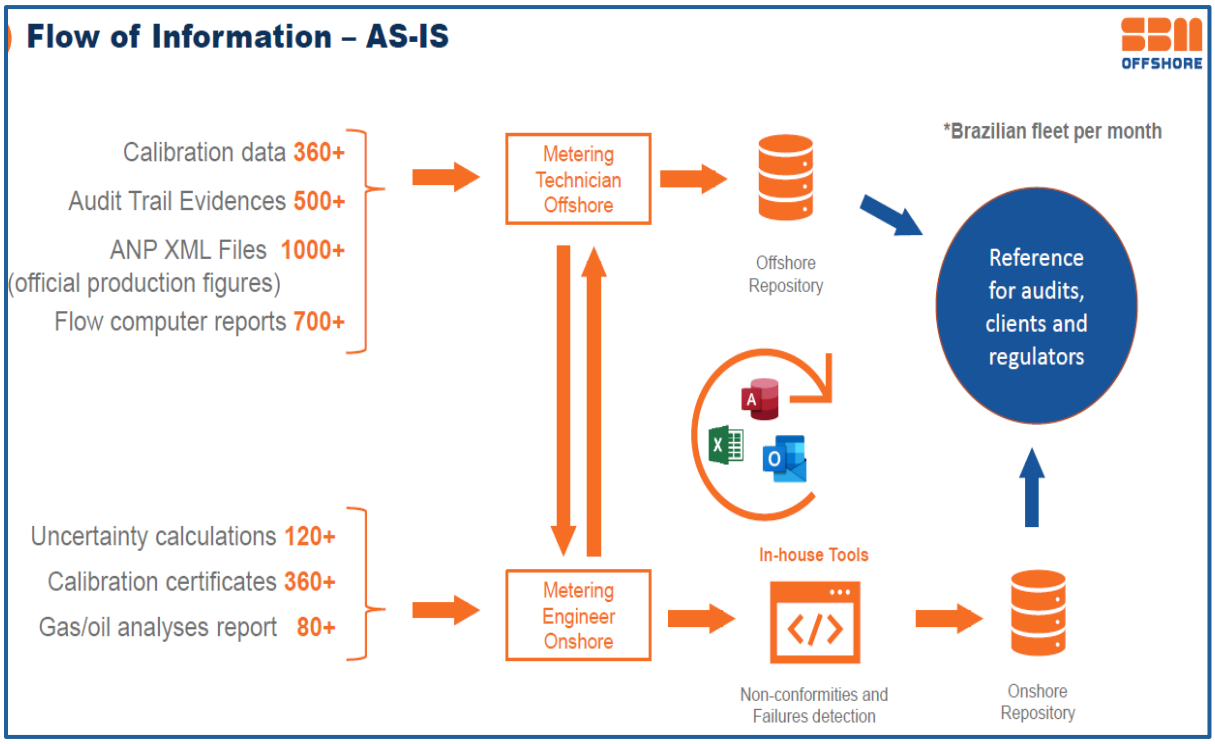

Figure 1 – Flow of Information – AS-IS

Key characteristics of the current state:

- **File-based**: Data is exchanged through shared folders and email, with no centralized database
- **Spreadsheet-driven**: Calibration schedules, sampling plans, and compliance tracking rely on Excel workbooks maintained manually
- **Fragmented tools**: In-house scripts and standalone utilities handle specific tasks (UC calculations, certificate generation) without integration
- **No audit trail**: Changes to data and configurations are not systematically logged
- **Manual consolidation**: Engineers spend significant time collecting, validating, and reformatting data from multiple sources

4.5 MANUAL EFFORT BASELINE

Interviews with operational teams and workload analysis established the following baseline for manual metering management activities:

Frequency	Effort per FPSO	Activities
Daily	~2.5 hours	Consolidating files, saving reports, generating certificates, validating FC configuration across 3 roles (ME, MA, MT)
Weekly	~3 hours	Compliance checks, schedule reviews, reporting
Monthly	~6 hours	ANP submissions, audit preparation, trend analysis
Annual	~1,100 hours	Total manual effort per FPSO

With 9+ FPSOs, Region 1 dedicates approximately **10,000+ hours per year** to manual metering management activities — capacity that could be reallocated to engineering analysis and proactive compliance.

4.6 REGULATORY EXPOSURE

Non-compliance with ANP metering regulations carries significant financial penalties:

Non-conformity	Fine per event (USD)
Serial number incorrectly changed	$30,000
Incorrectly reported data	$28,000
Incorrect flowmeter parametrization data	$225,000 – $250,000
Calibration completed after due date	$225,000 – $250,000
Calibration results implemented after due date	$225,000
Sampling completed after due date	$250,000
Sampling report issued after due date	$225,000
Sampling results implemented after due date	$250,000
Operate equipment beyond regulatory limits	$250,000

2025 Near-Misses

FPSO	Issue	Exposure (USD)
CDM	Gas composition updated 6 hours after deadline	$120,000 – $1,000,000
ATD	Flowmeter calibration curve update 3 days late	$225,000
SEP	Prover calibration results updated 3 days late	$225,000
CDS	Failure notification sent after 72-hour deadline	$30,000
Total		Up to $1,480,000

4.7 GAP SUMMARY

The current manual, fragmented approach creates the following critical gaps that the MMT must address:

1. **No centralized database:** Data is scattered across folders, spreadsheets, and local tools with no single source of truth
2. **No real-time SLA tracking**: Deadline compliance is monitored reactively, leading to near-misses and late submissions
3. **No automated critical analysis**: Certificate and lab report validation is performed manually, increasing error risk
4. **No integrated audit trail**: Regulatory audits require significant manual effort to reconstruct data history
5. **No proactive alerting**: Teams are unaware of approaching deadlines until they become urgent or are already missed
6. **No fleet-wide visibility**: Management lacks a consolidated view of compliance status across all FPSOs

5 OBJECTIVES

This section specifies the measurable, overarching goals of the MMT product. It links the tactical software delivery to the strategic outcomes expected by SBM Executive Leadership.

5.1 MAIN GOAL

Achieve zero ANP metering fines across the SBM Offshore Brazilian fleet by providing a centralized, regulation-aware digital platform that replaces manual metering management with automated, auditable workflows.

5.2 SMART OBJECTIVES

Each objective is traceable to the Business Value drivers defined in Section 6 and aligned with the PACE program strategy.

ID	Objective	Baseline	Target	Timeline
OBJ-1	Reduce regulatory fine exposure	~$2.5M/year avg	> 40% risk reduction	12 months post-deployment
OBJ-2	Reduce manual metering effort	~1,100 hrs/FPSO/year	> 40% reduction (< 660 hrs)	12 months post-deployment
OBJ-3	Ensure audit readiness	Days to produce evidence	< 1 hour (automated export)	Upon M7 delivery
OBJ-4	Provide real-time SLA visibility	0% automated alerting	100% deadlines with ≥ 48h early warning	Upon M6 delivery
OBJ-5	Centralize all metering data	Fragmented (folders, email)	100% in single platform	Full module deployment
OBJ-6	Scale to full fleet	N/A	12 FPSOs without code changes	M11 as prerequisite

5.3 PACE ALIGNMENT

The MMT initiative is positioned within SBM Offshore's PACE (Program for Agile and Continuous Excellence) roadmap as a digital solution that delivers measurable value across two dimensions:

PACE Dimension	OKR	Annual Value (Fleet)
Risk exposure reduction	> 40% reduction in regulatory fine risk	~$1,000,000/year avoided
Manpower reallocation	> 40% reduction in manual metering hours	+4,000 hours / +$450,000/year

6 BUSINESS VALUE & DRIVERS

This section identifies the primary business drivers motivating the MMT investment, detailing the specific Key Performance Indicators (KPIs) and phased value delivery expectations.

6.1 BUSINESS DRIVERS

ID	Driver	"So What" Outcome
BV-1	Regulatory compliance (ANP/INMETRO deadlines)	Fines of $28K–$250K per event avoided; $35.8M historical exposure eliminated
BV-2	Near-miss prevention	Shift from reactive to preventive; $1.48M 2025 exposure addressed
BV-3	Operational efficiency	+4,000 hrs / +$450K/year freed from manual work
BV-4	Audit readiness	Evidence preparation reduced from days to minutes
BV-5	Fleet scalability (3 new FPSOs)	New units onboarded via configuration, no custom development

6.2 KPIS

KPI	Baseline	Target	Owner
ANP fine exposure ($/year)	~$2.5M/year	> 40% reduction	Metering Lead
Manual metering hours (hrs/FPSO/year)	~1,100	< 660 (> 40% reduction)	Metering Lead
SLA deadlines with early warning	0%	100% at ≥ 48h	Metering Engineer
Audit evidence preparation time	Days	< 1 hour	Metering Analyst
FPSOs managed without code changes	6 (legacy)	12 (full fleet)	Product Owner
Data centralization coverage	0%	100% (certs, reports, UCs, FC configs)	Product Owner

6.3 VALUE BY RELEASE

Phase	Modules	Impact	Validation
Foundation	M11, M12	Digital hierarchy; role-based access; configurable parameters	12 FPSOs modeled; admin setup verified
Core Compliance	M1, M2, M3	SLA tracking; calibration lifecycle; equipment traceability	Zero missed deadlines (pilot); audit trail active
Visibility	M6, M8, M9	Proactive alerting; planning dashboard; NFSM tracking	100% SLA alerts; NFSM time reduced
Integration	M4, M5, M7, M10	IFS sync; ANP XML export; historical reports	Evidence in < 1h; XML validated

7 BUSINESS INTERFACES

This section defines the business-facing interfaces that provide inputs to or consume outputs from the MMT, detailing the integration methodologies and data flow logic.

7.1 IF-1 — IFS (CMMS)

Owner: IT / Maintenance Team
Business Event: Spare parts procurement and inventory synchronization linked to calibration planning (M2)
Data Flow:

- **Input to MMT:** Spare parts availability, procurement status, equipment onshore maintenance status.
- **Output from MMT:** Calibration-linked spare requirements.

Integration Method [MVP]: Scheduled background synchronization via REST APIs (polling every 12 hours) fetching JSON payloads.
Service Expectations: ME must be able to visually link spare procurement items from IFS to planned calibrations. The MMT APIs must gracefully handle IFS downtime during synchronization.

7.2 IF-2 — POWER BI

Owner: Metering Analyst (MA) / IT
Business Event: Management reporting, KPI dashboards
Data Flow:

- **Input from MMT:** Read-only replica database connection / REST API data extraction.
- **Output:** Three operational PBI reports: (1) Calibration KPI — lagging, (2) Calibration KPI — leading, (3) Chemical analysis KPI.

Integration Method [MVP]: Direct SQL views configured in the replica database to serve flattened data sets optimized for PowerBI ingestion (daily refresh).

Service Expectations: PBI native files (.pbix) delivered as part of the initial scope. Access control within the dashboards must follow the same FPSO-level RBAC rules as the web interface.

7.3 IF-3 — ANP (REGULATORY SUBMISSIONS)

Owner: Metering Analyst (MA)
Business Event: Regulatory data export (XML modules), failure notification submission (NFSM)
Data Flow:

- **Output from MMT:** Formatted XML files per Resolução ANP nº 65/2014, offloading XML per Ofício nº 906/2021, NFSM dynamic reports per Resolução nº 18/2014, and structured ZIP export packages.

Integration Method [MVP]: Manual export triggered by the user. The MMT automatically structures the files into the rigid folder hierarchy expected by ANP auditors.
Service Expectations: Complete avoidance of XML schema validation errors on the ANP portal. NFSM reports must be fully compiled and ready within 72 hours of failure detection. All raw data must be retained for ≥ 10 years.

7.4 IF-4 — THIRD-PARTY CERTIFICATE ISSUERS

Owner: Metering Engineer (ME)
Business Event: Reception of calibration certificates issued by external RBC/Inmetro accredited laboratories.
Data Flow:

- **Input to MMT:** XML payloads and/or PDF certificate files imported manually or via direct vendor API integration by the ME.
- **Output from MMT:** Automated Critical Analysis (CA) result; execution of Uncertainty Calculation (UC).

Integration Method [MVP]: File upload (XML/PDF) through the UI. MMT uses parsing engines to automatically extract variables.

Service Expectations: The parsing engine must successfully auto-populate certificate UI fields with >95% accuracy to reduce manual typing. CA is performed instantaneously upon upload.

7.5 IF-5 — LABORATORY (CHEMICAL ANALYSIS REPORTS)

Owner: Metering Engineer (ME)
Business Event: Reception of chemical analysis reports (CRO, PVT, Enxofre, Viscosity) from offshore or onshore labs.
Data Flow:

- **Input to MMT:** Digitized PDF lab reports uploaded by the ME.
- **Output from MMT:** Parsed molecular parameters; CA validation (Approved/Reproved); systematic generation of emergency sample triggers upon reproval.

Integration Method [MVP]: Drag-and-drop PDF upload with OCR/text parsing for standardized tabular Methane/Density/BSW values.
Service Expectations: Real-time feedback if PDF parsing fails. Extracted values are immediately subjected to 2-sigma deviation checks against historical baselines.

7.6 IF-6 — CLIENT (PETROBRAS, SHELL)

Owner: Metering Lead (ML)
Business Event: Contractual compliance reporting, rapid audit evidence sharing.
Data Flow:

- **Output from MMT:** Compliance dashboards, unified audit evidence packages (ZIP), calibration certificates, UC reports, and permanent hyperlinked URLs to stored documents.

Integration Method [MVP]: UI-based sharing using cryptographically secure URLs.
Service Expectations: External client users without SBM credentials must *not* access internal data unless explicitly shared via targeted links. Reports available natively in Portuguese and English.

7.7 IF-7 — MICROSOFT PLANNER (OPTIONAL / PHASE 2)

Owner: Product Owner / IT

Business Event: Unified task planning for calibrations, inspections, and samplings integrated into corporate workflows.

Data Flow:

- **Output from MMT:** Planned activities pushed natively to Microsoft Planner.
- **Input to MMT:** Task completion status (bidirectional sync).

Integration Method: Microsoft Graph API (OAuth2 authentication).

Service Expectations: Pushed to Phase 2, subject to SBM architectural approval. If adopted, replaces the dedicated M8 Planning UI.

8 BUSINESS CHALLENGES & INEFFICIENCIES

The current metering management approach relies on manual processes, fragmented repositories, and spreadsheet-based tracking. This creates significant operational risk on a fleet of 9+ FPSOs handling 2,900 regulated instruments and over 1 million critical parameters daily. The challenges below represent the primary pain points that justify the need for the MMT and guide feature prioritization.

ID	Challenge	Impact
CH-1	No automated deadline tracking for calibrations, sampling, and FC updates	$1.48M near-miss exposure in 2025; $2.5M/year avg in historical fines
CH-2	Manual data consolidation across offshore/onshore	~1,100 hrs/FPSO/year of low-value work
CH-3	Fragmented data across shared folders, email, and local tools	No single source of truth; data inconsistencies
CH-4	No systematic audit trail for configuration and status changes	Audit evidence takes days to reconstruct
CH-5	No automated Critical Analysis for certificates and lab reports	Risk of undetected anomalies; delayed reproval
CH-6	Equipment traceability gaps (serial numbers, tags, certificates)	Compliance risk during ANP audits
CH-7	No scalable model for onboarding new FPSOs	Each new unit duplicates manual processes
CH-8	No permanent hyperlinks or bilingual document support	Manual prep for client/regulator evidence sharing

9 DATA SOURCES & DEPENDENCIES

The MMT ingests data from multiple sources across offshore operations, onshore engineering, third-party laboratories, and enterprise systems. Early visibility of these dependencies is essential to avoid late-stage blockers during integration.

Most data sources (DS-1 through DS-4, DS-6) are currently available but maintained in spreadsheets, PDF/XML files, and local repositories. Integration into the MMT will primarily occur via manual entry or file synchronization from Cognite during initial phases, with API-based synchronization (particularly for IFS) to be defined during Framing & Design.

DS-1 (FPSO hierarchy) and DS-2 (equipment master data) represent the largest initial migration effort. A structured onboarding plan per FPSO should be defined to ensure data quality and completeness before go-live. DS-7 (holiday calendars) is required for business-day SLA calculations and must be configured per FPSO by admin.

ID	Data Source	Owner / SME	Criticality
DS-1	FPSO hierarchy & metering system configuration	ME / ML	High
DS-2	Equipment master data (serial numbers, tags, specs)	MT / ME	High
DS-3	Calibration certificates & UC reports (PDF/XML)	ME	High
DS-4	Chemical analysis lab reports (CRO, PVT, Enxofre, Viscosity)	ME	High
DS-5	Spare parts & procurement status (IFS)	IT / Maintenance	Medium
DS-6	Flow computer parameters & readings	MT	High
DS-7	Holiday calendars (per FPSO / region)	ML / Admin	Medium

10 ASSUMPTIONS & DEPENDENCIES

10.1 ASSUMPTIONS

The following assumptions underpin the requirements in this document. If any proves invalid, impacted requirements must be reassessed.

1. **Data quality**: Metering data currently maintained in spreadsheets and local files is accurate and complete enough to serve as the baseline for initial migration into the MMT.
2. **SME availability**: The SBM Metering Team (ME, MA, MT) will be available as subject matter experts throughout Framing, Design, and UAT phases despite high operational workload.
3. **Regulatory stability**: ANP regulatory requirements (RTM 01/2013, Resolução 18/2014, Resolução 52/2013) will remain stable during the development period. Changes will be handled via the change control process.
4. **Common hierarchy model**: All FPSOs share a common metering hierarchy structure (FPSO → Metering System → Meter Run → Instruments) that can be modeled through a single configurable data model.
5. **Offshore connectivity**: Internet connectivity on offshore FPSOs is sufficient for web-based access to the MMT, at least for data entry and status updates by Metering Technicians.
6. **Lab report formats**: Third-party laboratories will continue delivering calibration certificates and analysis reports in PDF and/or XML formats compatible with MMT import capabilities.
7. **Holiday calendars**: Holiday calendars required for business-day SLA calculations will be provided and maintained by SBM Operations on a per-FPSO/region basis.

10.2 EXTERNAL DEPENDENCIES

ID	Dependency	Risk if Delayed
D-1	IFS (CMMS) API access for spare parts integration	M2 planning limited to manual spare tracking
D-2	Product Owner assignment with metering domain knowledge	Requirement ambiguity; delayed acceptance
D-3	Dedicated engineer (100%) with metering background	Knowledge gaps; increased rework cycles
D-4	ANP XML schema specifications for export validation	M7 export cannot be validated against regulatory format
D-5	SBM IT infrastructure (hosting, database, network)	Deployment timeline at risk

11 CONSTRAINTS

The following limiting factors must be respected during product design and delivery.

11.1 REGULATORY

1. **ANP compliance**: The MMT must comply with all reference documents listed in Section 2, particularly RTM 01/2013, Resolução 18/2014, Resolução 52/2013, and Resolução 65/2014. Non-compliance is not acceptable.
2. **Data retention**: All data (certificates, reports, production data, audit trail) must be stored and accessible for a minimum of 10 years.
3. **NFSM timelines**: Failure notifications must support submission within 72 hours of detection, as mandated by Resolução ANP nº 18/2014.

11.2 TECHNICAL

4. **No hardcoded parameters**: All parameters used in the software (instrument types, meter classifications, calibration frequencies, SLA intervals, spare locations, etc.) must be configurable by admin users, even where not explicitly specified in the Functional Specification.
5. **Bilingual documents**: All documents generated by the MMT (reports, certificates, etc.) must be available in both Portuguese and English, either as two separate documents or ideally as a single bilingual document. Templates subject to SBM approval.
6. **English UI**: The software interface language must be English.
7. **Permanent hyperlinks**: Every attachment, certificate, UC, or report stored in the MMT must have a permanent hyperlink that can be shared with other users, respecting access control rules.
8. **API-first architecture**: All functionalities must be exposed via authenticated REST APIs, enabling data access by SBM users and external integrations (Power BI, IFS).

11.3 OPERATIONAL

9. **Filter persistence**: Applied filters on any page must be stored per user and restored on subsequent access. A clear filter button must be available, and the cleared state must also be persisted.

10. **Shareable URLs**: Each page must have its own unique link. Applied filters must be reflected in the URL so that shared links reproduce the exact same view for other authorized users.
11. **Template approval**: All document templates (certificates, reports, checklists) are subject to SBM approval prior to deployment.

11.4 CONTRACTUAL

12. **Support period**: The contractor must provide 2 years of technical support and bug fixes after delivery.
13. **Active development**: 1 year of active development after delivery for functionality fixes.
14. **Required deliverables**: The contractor must deliver: detailed SRS, periodic project reports, project timeline with deployment and UAT dates, training material, and user manual.

12 BUSINESS & FUNCTIONAL REQUIREMENTS

This section specifies what the MMT must do to meet business needs, organized by module. Requirements are written as structured statements ("The system shall...") and grouped by capability.

Detailed rules reflect both the original Functional Specification and the implemented MVP capabilities (including data extracted from *All Periodic Analyses* SLA tables). Requirements marked with **[MVP]** have been validated in the current development baseline.

The MVP source code is hosted on GitHub Repository: **https://github.com/viniciosgnr/MMT** and deployed on Versel domain: **https://mmt2-omega.vercel.app/**.

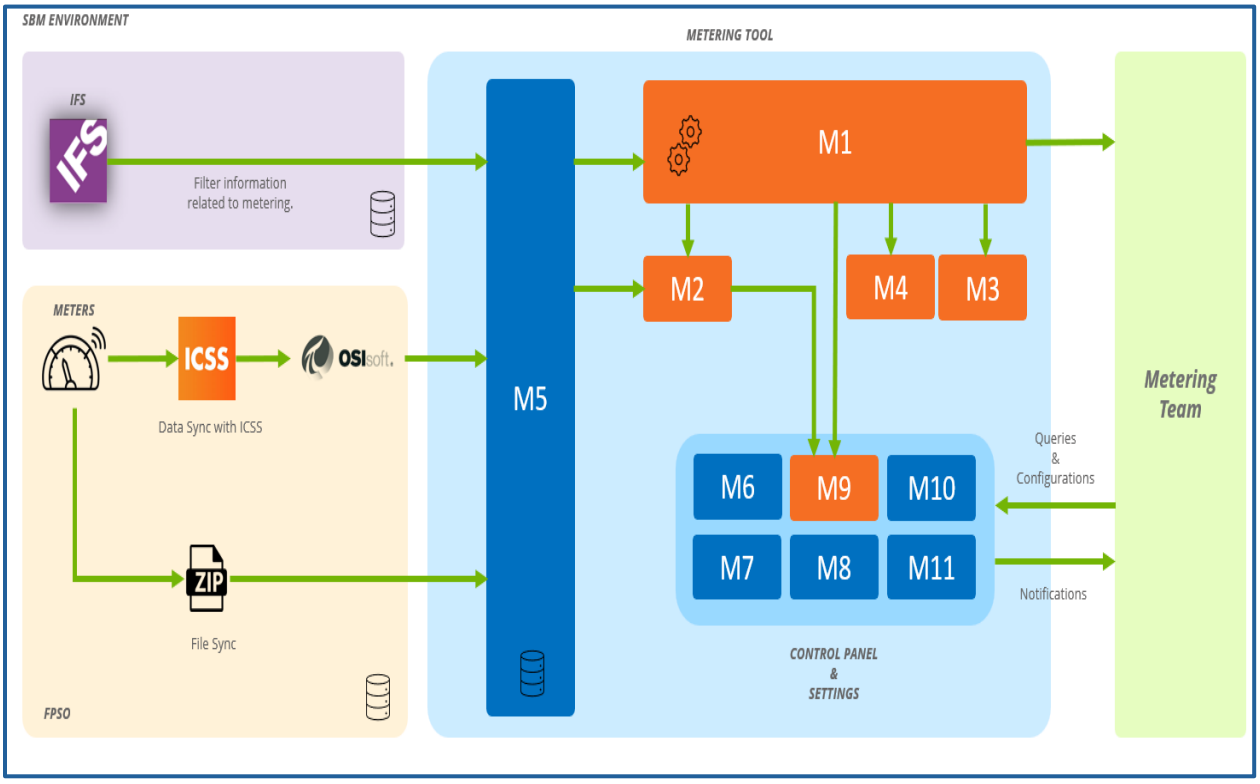

Figure 2 – High-Level Architecture Expected

12.1 CONFIGURATION MODULE (M11)

Centralizes all parametrization used across the software, ensuring no parameter is hardcoded.

1. The system shall allow admin users to model the complete hierarchical mapping (FPSO → Metering System → Meter Run → Instruments) and assign specific regulatory classifications (e.g., Custody Transfer, Allocation). **[MVP]**
2. The system shall support a dynamic property engine, allowing admins to define custom fields for any hierarchy level or equipment type with explicit validation rules (min/max numeric ranges, mandatory flags, datatypes, picklists). **[MVP]**
3. The system shall allow admins to independently create, update, and remove Equipment Types and their associated dynamic property schemas. **[MVP]**
4. The system shall allow configuration of calibration and sampling frequencies specific to each FPSO and system classification, handling operational overlaps gracefully (the lowest interval applies). **[MVP]**

5. The system shall support the configuration of a multi-dimensional SLA Matrix (Classification × Analysis Type × Location) defining explicit target days for each workflow phase. **[MVP]**
6. The system shall manage external logistics parameters, including holiday calendars, shift patterns, and onshore warehouse locations to calculate true business days accurately.

12.2 EQUIPMENT REGISTRATION & INSTALLATION (M1)

Manages serial numbers, tags (functional locations), and the complete lifecycle of physical movement.

1. The system shall allow users to register physical Serial Numbers, dynamically enforcing the property schemas defined in M11 for the selected Equipment Type. **[MVP]**
2. The system shall manage Tags as fixed hierarchical addresses tied to P&IDs, supporting systematic equipment swap operations. **[MVP]**
3. The system shall track the complete installation history of every serial number against tags, maintaining an immutable audit trail of installation and removal dates. **[MVP]**
4. The system shall maintain an aggregated historical view of all certificates, maintenance reports, and calibrations related to both the physical device (serial number) and the location (tag) over time. **[MVP]**
5. The system shall dynamically filter valid tags for a serial number during installation based on metadata compatibility (e.g., Fluid Type or System matching). **[MVP]**
6. The system shall auto-generate the formatted Equipment Change Report (Appendix D layout) with a mandatory checklist upon any installation or replacement action.

12.3 METROLOGICAL CONFIRMATION (M2)

Manages the full calibration lifecycle per ISO 10012:2003 and Resolução ANP nº 18/2014.

1. The system shall track calibrations through a strict, auditable workflow: Planning → Execution → Seal Control → Certificate Issuance → Critical Analysis (CA) → Uncertainty Calculation (UC) → Flow Computer (FC) Update → Closure.
2. The system shall calculate temporary Flow Computer override values per ANP regulations when a meter is removed from an operating line for ex-situ calibration.
3. The system shall manage physical seal tracking (installation, removal, current status per tag), enabling generation of targeted seal reports for regulatory audits.
4. The system shall support automated extraction of parameters from imported calibration certificates (XML/PDF formats) issued by 3rd-party vendors (e.g., Inmetro/RBC labs).
5. The system shall perform automated Critical Analysis (CA) using historical 2-sigma deviation checks and hard-limit rules to explicitly flag certificates as Approved or Reproved.
6. The system shall execute on-demand and scheduled Uncertainty Calculations (UC) utilizing the latest approved parameters.
7. The system shall clearly visually flag or watermark any generated documentation as "FOR REFERENCE ONLY" until the final FC Update operational step is formally closed and approved.

12.4 CHEMICAL ANALYSIS (M3)

Manages the sampling workflow from physical collection through flow computer update.

1. The system shall support diverse sampling procedures: Well Test Oil, Daily Oil Sample (BSW/Density), and Periodic Gas/Oil Chromatography (CRO/PVT). **[MVP]**
2. The system shall track samples through up to 11 dynamic workflow steps (Plan → Sample → Logistics to Shore → Warehouse → Lab Delivery → Report Issue → CA Validation → FC Update). **[MVP]**
3. The system shall dynamically bypass irrelevant workflow steps contextually (e.g., skipping transport logistics for offshore analyses, or bypassing the FC Update step for Sulfur/Viscosity analyses that only inform limits). **[MVP]**
4. The system shall evaluate and enforce target SLA deadlines at every workflow step by querying the SLA matrix (M11), calculating durations in business days based on location and classification. **[MVP]**

5. The system shall automatically ingest and parse PDF lab reports, extracting complex compositional data (e.g., Methane fraction, Density, O2) and verifying them against regulatory limits. **[MVP]**
6. The system shall trigger an Emergency Sample procedure upon Critical Analysis reproval, auto-scheduling it for the earlier of: 3 business days post-emission or the next periodic date. **[MVP]**
7. The system shall systematically auto-calculate and schedule the next planned periodic sample date immediately upon completion of the current cycle. **[MVP]**

12.5 ONSHORE MAINTENANCE (M4)

Tracks equipment maintenance at onshore facilities.

1. The system shall manage maintenance activity logs for equipment returned to the designated onshore warehouse.
2. The system shall provide structured integration linkage to IFS (CMMS) Work Orders for tracking repair status.
3. The system shall explicitly block the re-installation of any equipment to a Custody Transfer (CT) tag if the equipment is flagged as "under maintenance."

12.6 SYNCHRONIZE DATA (M5)

Handles data exchange with external systems via structured APIs.

1. The system shall expose secure, documented REST APIs for external systems to fetch approved calibration parameters and evaluated lab reports. **[MVP]**
2. The system shall fetch CMMS (IFS) spare parts inventory and equipment status incrementally or via scheduled background sync. **[MVP]**
3. The system shall monitor synchronization health continuously and permanently flag failed API payloads in an administrative dashboard for review.

12.7 MONITORING & ALERTS (M6)

Provides proactive alerting for SLA compliance and parameter deviations.

1. The system shall continuously monitor Flow Computer configurations against the latest approved MMT Certificates and Lab Reports, raising an explicit system alarm if the FC operates out of compliance. **[MVP]**
2. The system shall generate automatic alerts for upcoming and overdue expirations on Calibration Certificates and Periodic Sampling, escalating warnings functionally (e.g., 30 days, 15 days, day of, overdue).
3. The system shall identify anomalies in daily production parameters (e.g., unchanged daily volumes or static temperatures) using standard deviation rules.
4. The system shall trigger system-level SLA breach alerts across any of the M3 (Sampling) or M2 (Calibration) workflow steps immediately if the expected completion date passes.

12.8 EXPORT DATA (M7)

Exports structured data for auditing purposes.

1. The system shall export filtered module views (e.g., equipment lists, certificates, activities) to Excel (.xlsx) and PDF formats. **[MVP]**
2. The system shall generate a structured ZIP archive for ANP audits, strictly following the folder structure: FPSO → Category → Tag → Date_EventCode.
3. The system shall dynamically package all associated metadata, external XML files, and PDF certificates in the generated ZIP archives.

12.9 PLANNING (M8)

Provides a unified view of all outstanding activities.

1. The system shall provide aggregated calendar and list views of all planned calibrations, inspections, and sampling activities across the fleet. **[MVP]**
2. The system shall use a traffic-light color coding system determined by the SLA threshold configurations in M11 (overdue, impending, on-track).

3. The system shall allow users to log mitigations for overdue activities, attaching justifications and establishing a new projected target date without losing the original SLA breach record.
4. The system shall allow filtering of activities by FPSO, time period, and activity type (e.g., plan calibration, execute calibration, take sample, issue report).

12.10 FAILURE NOTIFICATION — NFSM (M9)

Manages mismeasurement notifications per Resolução ANP nº 18/2014.

1. The system shall manage the complete lifecycle (opening, editing, approving, closing) of Failure Notifications (NFSM).
2. The system shall differentiate between Initial Notifications (which must be submitted ≤ 240 hours from detection) and Final Notifications.
3. The system shall mandate explicit Metering Engineer (ME) approval for the closure of any NFSM.
4. The system shall structure captured NFSM data in a format completely compliant with the ANP XML schema, facilitating immediate export for the portal.
5. The system shall support configurable email distribution lists per FPSO to automatically dispatch PDF reports of approved NFSMs to stakeholders.

12.11 HISTORICAL REPORTS (M10)

Stores ad-hoc reports not covered by other modules.

1. The system shall serve as an auditable repository for unstructured but operationally relevant documents (e.g., general ANP Letters, Offloading logs, daily HMI files). **[MVP]**
2. The system shall allow users to optionally link uploaded files to existing functional nodes (FPSO, Metering System, Tag) and physical Serial Numbers for contextual retrieval.
3. The system shall support bulk uploading of multiple documents concurrently.
4. The system shall automatically preserve upload timestamps and usernames to guarantee non-repudiation.

12.12 USER & ACCESS MANAGEMENT (M12)

Controls authentication, authorization, and user administration.

1. The system shall support Role-Based Access Control (RBAC) with predefined hierarchical access profiles (e.g., System Admin, Metering Engineer, Metering Technician).
2. The system shall enforce rigorous tenant-like isolation at the FPSO level: users shall only access and view data for their explicitly assigned vessels.
3. The system shall log every mutating user action (Create/Update/Delete) across all modules in an immutable Audit Log.
4. The system shall extend all Role-Based Access Control rules to the underlying REST APIs, ensuring the exact same permissions are evaluated programmatically.

13 NON-FUNCTIONAL REQUIREMENTS

This section defines the quality attributes, performance goals, and operational limits that the MMT must meet to be successful, combining explicit constraints from the Functional Specification with standard SaaS baseline requirements.

13.1 USABILITY & ACCESSIBILITY

1. The software shall be entirely web-based, accessible via modern desktop and mobile browsers (e.g., Chrome, Edge) without requiring supplemental local installation.
2. The user interface must proactively group related tasks to minimize clicks and screen transitions, enforcing an intuitive logical flow.
3. System interface and displays must be exclusively in English, but dynamically generated documents (reports, certificates) must support Portuguese/English bilingual modes to satisfy local regulatory bodies and international stakeholders.

13.2 PERFORMANCE AND SCALABILITY

1. **Page Load:** The system shall ensure no single screen takes longer than 10 seconds to fully load under standard network operations.
2. **Workflow Speed:** Key operational workflows, specifically registering a calibration and issuing a related Uncertainty Calculation report, must be achievable within 1 minute by a trained user.

3. **Concurrency:** The backend architecture must reliably support at least 10 simultaneous active power users interacting with the database without performance degradation.
4. **Data Scale:** The database must scale gracefully to accommodate at least 15 FPSOs along with 10 years of accumulated hierarchical telemetry and documentation history.

13.3 SECURITY & AUTHORIZATION

1. **Data in Transit:** All traffic between the client and backend APIs must be encrypted using HTTPS/TLS 1.2+.
2. **Authentication:** The system shall authenticate users via secure login/password mechanisms, supporting password recovery.
3. **Tenant Isolation:** The application logic must strictly enforce FPSO-level authorization isolation. Users shall not view or modify data pertaining to assets outside their assigned permissions.
4. **Immutability:** The Audit Log shall be immutable and non-repudiable.

13.4 AVAILABILITY & RELIABILITY

1. **Uptime:** The application shall target 99.9% availability during business hours to support critical regulatory and operational routines.
2. **Failures:** The system shall handle API integration failures (e.g., IFS CMMS sync timeouts) gracefully, logging the failure without crashing or blocking the user interface.
3. **Backup Recovery:** Automated daily backups of the database and object storage (uploaded certificates/pdfs) shall be securely maintained.

13.5 DEPLOYMENT STRATEGY

1. **Phased Rollout:** To mitigate risk, the MMT shall be deployed incrementally. Each phase will introduce a standalone module.
2. **Sequence Requirements:** Deployment serialization must respect technical dependencies. The strict initial sequence is: *Configuration → Managing Equipment*. Subsequent suggested order: *Chemical Analysis → Planning → Export Data → Sync Data → Alerts → Metrological Confirmation → NFSM → Historical Reports → Maintenance*.
3. **Continuous Feedback:** A formal period for user testing, feedback aggregation, and bug resolution must be established before progressing to the subsequent deployment phase.

13.6 TESTING & UAT

1. **Formal UAT:** Every software deliverable shall undergo formal User Acceptance Testing (UAT) executed by designated SBM users.

2. **UAT Documentation:** The contractor is required to furnish a comprehensive UAT Document per phase, containing explicit test cases and structured feedback forms.
3. **Completion Criteria:** A consolidated UAT Report aggregating all feedback must be issued to formalize module acceptance.

13.7 DATA RETENTION & RETRIEVAL

1. **Auditable Lifespan:** To satisfy stringent ANP auditing rules, all system data, metadata, and attached files must be retained and accessible for a minimum of 10 years.
2. **Permalink Strategy:** Every generated and uploaded document within the MMT must feature a permanent, shareable hyperlink that enforces RBAC upon access.
3. **Filter Persistence & Sharing:** The exact state of any data table filter must be both persisted for the user session and serialized into the URL, allowing exact-view sharing between authorized users.

14 REQUIREMENTS TRACEABILITY MATRIX (RTM)

This section maps the Business Objectives (defined in Section 5) to the explicit Functional Modules (defined in Section 12). This matrix ensures that every technical feature developed has a direct linkage to a business outcome, facilitating value-driven UAT validation and scope management.

Business Objective	Target Outcome	Primary Modules Enablers	Technical Capability Linkage
OBJ-1: Reduce administrative effort	80% reduction in manual data entry and consolidation	M3, M5, M7	Automated PDF parsing for lab reports (M3). Background JSON API sync with IFS (M5). Automated one-click ZIP generation for ANP audits (M7).
OBJ-2: Eliminate regulatory fines	$0 fines related to missed deadlines / late notifications	M6, M8, M9	Proactive configurable SLA alerts and parameter deviation alarms (M6). Traffic-light fleet planning dashboard (M8). Automated XML formatting and dispatch of structured NFSMs (M9).
OBJ-3: 100% Data traceability	Audit-ready historical log for any Flow Computer parameter	M1, M2, M12	Immutable audit logs for all mutating actions (M12). Historical tag vs. serial number tracking (M1). Persistent "FOR REFERENCE ONLY" state for unapproved parameters (M2).

OBJ-4: Standardize fleet handling	100% of FPSOs on a single logic tree	M11, M1	Centralized configuration of a dynamic property engine enforcing rules across all units (M11). Semantic metadata compatibility filtering during installation (M1).
OBJ-5: Accelerate Critical Analysis (CA)	60% faster approval cycle for certificates and labs	M2, M3	Automated 2-sigma deviation checks against historical baselines immediately upon XML/PDF upload (M2, M3).
OBJ-6: Single Source of Truth	Centralized, accessible data lake for all meter data	M10, M12, IF-2	Hyperlinked, 10-year retained repository of unstructured records (M10). Direct integration serving flat SQL views to PowerBI (IF-2). Rigid FPSO-level RBAC isolation (M12).

15 APPENDIX

APPENDIX A **MVP MOCKUP SCREENS**

APPENDIX B **PVT LAB REPORT TEMPLATE**

APPENDIX C **CRO LAB REPORT TEMPLATE**

APPENDIX D **EQUIPMENT CHANGE REPORT**

APPENDIX D **CHEMICAL ANALYSIS SLA MATRIX**